Physical Chemistry Cume

Rotational and Vibrational Spectroscopy

April 15th 2011

Constants and conversion factors:

The mass of ^{1}H is 1.0079 a.m.u, the mass of ^{35}Cl is 34.9688 a.m.u and the mass of D (^{2}H) is 2.016 a.m.u (a.m.u. stands for atomic mass units).

1 a.m.u. = 1.661×10^{-27} kg

c = 2.998×10^{8} ms^{-1}

h = 6.626×10^{-34} Js

Some useful equations:

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\nu = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$$

For a diatomic molecule: $I = \mu r^2$

$$\widetilde{F}(J) = \widetilde{B}J(J+1)$$

$$\widetilde{B} = \frac{h}{8\pi^2 cI}$$

Where $\widetilde{F}(J)$ and $\widetilde{B}$ denote that the units are in wavenumbers

1) For the following 8 molecules
   (i) CH_4 , (ii) CH_3Cl , (iii) N_2 , (iv) C_6H_6 (benzene), (v) CO_2, (vi) CO, (vii) ^{1}H-^{35}Cl, (viii) D-^{35}Cl

   a) How many normal modes of vibration does each molecule have?
   b) Which molecules would show IR activity (i.e. absorb some infrared light)?
   c) Which molecules would be microwave active?
   d) Which molecules would be Raman active?

   (40 points)

2) Rotational (microwave) spectroscopy is a very important tool for measuring bond lengths of molecules. For ^{1}H-^{35}Cl, the microwave spectrum shows a series of peaks separated by 20.8784 cm^{-1}. For D-^{35}Cl (where D is deuterium, ^{2}H), the microwave spectrum shows a series of peaks separated by 10.7840 cm^{-1}.

   Assuming a rigid rotor model, what are the lengths of the ^{1}H-^{35}Cl and the D-^{35}Cl bonds? Briefly explain the comparison of these bond lengths.

   (30 points)

3) For ^{1}H-^{35}Cl, the fundamental vibrational frequency is 2988 cm^{-1}. What would be the fundamental vibrational frequency (in cm^{-1}) of D-^{35}Cl? What assumptions do you make?
   (30 points)